

Research Statement

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1 Introduction

The MM^{++} *Problem* is one of the holy grails of set theory, motivating the development of several fields of research. I would like to contribute to its resolution during my carrier.

Problem 1.1 (MM^{++} Problem). *What is the exact consistency strength of Martin's Maximum⁺⁺ (MM^{++})?* $\square$

The study of lower and upper bounds on the consistency strength of MM^{++} has given rise to multiple sub-branches of set theory. Let me start from the lower bound problem. The following question has been the most promising way so far to approach it.¹

Problem 1.2 (Descriptive Inner Model Problem). *Let ϕ be a large cardinal notion (expected to be weaker than MM^{++}). Assuming MM^{++} , is there a forcing extension and within it, a pointclass $\Delta \subseteq \mathcal{P}(\mathbb{R})$ such that $L(\mathbb{R}, \Delta) \models \text{AD}^+$ and $\text{HOD}^{L(\mathbb{R}, \Delta)} \models \phi$?* $\square$

The standard technique used for answering this question is the *Core Model Induction*. If ϕ is the statement “there exists a Woodin cardinal which is also a limit of Woodin cardinals”, the necessary machinery underlying the Core Model Induction is already available, making this a problem I believe I could resolve in medium-term. If ϕ is the statement “there exists a measurable Woodin cardinal”, we have an idea what the machinery looks like, but we do not know that it works as expected. This gives rise to a project that I would like to reflect on in medium-to long-term. On the other hand, if ϕ is the statement “there exists a beth-fixed point λ and a λ -supercompact cardinal”, we are in uncharted territories.²

The underlying machinery for the Core Model Induction requires verification of some form of the *Mouse Set Conjecture*.

¹This approach was used in [ST23] to obtain the best known lower bound on the consistency strength of MM^{++} .

²In this region, the crucial notion of a *mouse* has not been successfully defined. Among other things, this makes it impossible to precisely state the *Mouse Set Conjecture* mentioned in the continuation.

Problem 1.3 (Mouse Set Conjecture). *Let ψ be an anti-complexity notion. Assuming $\text{AD}^+ + \psi$, does Mouse Capturing³ hold?* $\square$

This is a problem that greatly interests me.⁴ The way I stated it, it depends on an anti-complexity notion ψ . When approaching the Descriptive Inner Model Problem for a Woodin cardinal which is a limit of Woodin cardinals, the relevant anti-complexity assumption is the statement “there is no strategy for a mouse with a Woodin cardinal which is a limit of Woodin cardinals”. A variant of the Mouse Set Conjecture in this case is known to be true.⁵ On the other hand, if the large cardinal notion “Woodin cardinal which is a limit of Woodin cardinals” is replaced with “measurable Woodin cardinal”, we arrive to a form of Mouse Set Conjecture of at least medium-term difficulty.

Let me now say a few words on the upper bound problem for the consistency strength of MM^{++} . The best known upper bound is obtained through a forcing iteration of semi-proper posets.⁶ This iteration is done with the aim of saturating the universe with partially generic filters for semi-proper posets. It turns out that once this is achieved, the universe is saturated for partially generic filters for posets in a larger class, the class of stationary-set-preserving (ssp) posets. This saturation is exactly what MM^{++} is. On the other hand, a direct attempt to forcing MM^{++} would be to try iterating ssp posets themselves. The problem is that there is no general iteration theorem for ssp posets and there are actually examples of when the iteration is not possible.⁷

Problem 1.4 (SSP Iteration Problem). *Characterize when it is possible to iterate ssp posets and what that iteration is.* $\square$

There is no satisfying answer to this question when working with assumptions of consistency strength below a supercompact cardinal.⁸ I believe I can bring insights in this direction in medium-term; for example, some ideas in my previous work appear promising.⁹ Good test questions are restricted forms of MM^{++} , such as the *Axiom* (*).¹⁰

Problem 1.5. *Assuming there exists a proper class of Woodin cardinals, can we (provably) force (*)?* $\square$

³[Sar15, page 1]

⁴The fact that it has the positive answer for a certain ϕ is used prominently in the paper [KS24] of Sargsyan and me.

⁵Sargsyan showed in [Sar23] and [Sar] that a variant of Mouse Capturing, called the *Hod Pair Capturing* (vide [Ste23b, Definition 1.7.1]), is true in this case. The Hod Pair Capturing follows from Mouse Capturing, but it is open whether they are equivalent.

⁶[FMS88]

⁷The book [She98] provides an example of a length ω iteration of ssp forcings that adds a descending sequence of functions above all canonical functions (ordered modulo a club). This sequence of posets cannot be extended to a length $\omega + 1$ iteration without collapsing ω_1 .

⁸It was shown in [She98] that a certain reflection principle implies that every ssp poset is semiproper. The same source develops an iteration theory for semiproper posets. The two results combined effectively solve the SSP Iteration Problem, provided the said principle holds. However, the best known upper bound on the consistency strength of that principle is a supercompact cardinal, which does not help with the goal of reducing the upper bound on consistency strength of MM^{++} .

⁹[KV23] ¹⁰[AS21]

The case of this principle is curious because it is known that its consistency strength is below a proper class of Woodin cardinals,¹¹ but we know how to force it only by assuming a supercompact cardinal.¹² The better bounds on the consistency strength are obtained by forcing over a model of the *Axiom of Determinacy*⁺ (AD^+). This axiom contradicts the Axiom of Choice, so the question here is whether the arguments under AD^+ can be “transformed” into arguments in ZFC . An abstract principle that would allow such a transformation is *the completeness for Ω -logic*. If there exists a proper class of Woodin cardinals and the Ω -logic is complete, then there is a poset that forces $(*)$.

Problem 1.6 (Ω -conjecture). *Is it true that large cardinals imply that Ω -logic is complete?*¹³ $\square$

In the minimal model with a proper class of Woodin cardinals Ω -conjecture is true, so we know that there is a poset forcing $(*)$. This answer is unsatisfactory for two reasons. First, much more is assumed on the universe in addition to large cardinals, making this result very far from the *provable* forcibility asked in Problem 1.5. Second, the poset obtained through the application of the completeness of Ω -logic is most likely not ssp. Even though this was not asked for explicitly in Problem 1.5, it is what we would need to get insight into the SSP Iteration Problem. This relates to the following conjecture.

Problem 1.7 (Σ_2 Forcibility Conjecture). *Is it true that large cardinals imply that every Σ_2 formula that can be forced, can be forced by an ssp poset?*¹⁴ $\square$

The conjecture is known to fail in some cases where restrictive conditions are imposed on the universe.¹⁵ If it assumed that there exists a proper class of Woodin cardinals which are also limits of Woodin cardinals, there are no known counter-exmpales to the conjecture. I find Ω -conjecture and Σ_2 Forcibility to be very motivating problems, but in their full generality, they are most likely long-term projects.

I will now elaborate on my previous work, relate it to the questions I have raised so far, and point to other questions that the work naturally incites.

2 Building Models of Determinacy from Below

The fact that the Wadge reducibility is a pre-well-order leads to the natural question of whether it is possible to describe the next determined set in the Wadge hierarchy, provided that we are given the previous ones. For example,

¹¹[Woo99]

¹²Viz. we force the full MM^{++} .

¹³[Woo99, Chapter 11, Question 23]

¹⁴This conjecture is alluded to in [Woo99, Section 10.1].

¹⁵Since Ω -conjecture is true in mice and since the statement “the non-stationary ideal on ω_1 is dense” can be forced over $\text{ZF} + \text{AD} + V = L(\mathbb{R})$ (vide [Woo99]), it follows that this statement can be forced over the minimal mouse with a proper class of Woodin cardinals and an inaccessible limit of Woodin cardinals. On the other hand, an unpublished argument due to Woodin shows that any such forcing must collapse ω_1 .

the next determined set above all sets of reals in $L(\mathbb{R})$ is essentially $\mathbb{R}^\sharp$: if there exists a model of AD^+ extending $L(\mathbb{R})$ and having more sets of reals, then that model contains $\mathbb{R}^\sharp$. The answer to the question is negative in general since the existence of *divergent models* of AD^+ is consistent.¹⁶ Nevertheless, a description of this kind was given by Steel and Woodin for sets of reals appearing in the minimal model of $\text{AD}_{\mathbb{R}} + \text{DC} +$ “the quasi-club filter on $[\mathcal{P}(\mathbb{R})]^\omega$ is a supercompact measure”.¹⁷ Sargsyan and I extended this result to the minimal model of $\text{AD}_{\mathbb{R}} +$ “ Θ is regular”, and to somewhat larger models.

More precisely, we described an L -like construction of the minimal model of $\text{AD}_{\mathbb{R}} +$ “ Θ is regular”. The focus of the construction are *determinacy pointclasses*, i.e. collections Δ of sets of reals such that $\mathcal{P}(\mathbb{R}) \cap L(\mathbb{R}, \Delta) = \Delta$ and $L(\mathbb{R}, \Delta) \models \text{AD}^+$. The first thing to do is to define an operation $\mathcal{O}$ that takes a determinacy pointclass and returns a bigger one. For any determinacy pointclass Δ and any ordinal κ , we let $\mathcal{O}(\Delta, \kappa)$ be the pointclass

$$\mathcal{P}(\mathbb{R}) \cap L(\Delta^\omega, \kappa^\omega)[\mu_\Delta, \mu_\kappa],$$

where, for any set X , μ_X denotes the club filter on $[X]^\omega$. If there exists κ such that

$$\Delta \subsetneq \mathcal{O}(\Delta, \kappa),$$

we let κ_Δ be the smallest such κ , otherwise κ_Δ is undefined. Finally, $\mathcal{O}(\Delta)$ is defined to be $\mathcal{O}(\Delta, \kappa_\Delta)$, if κ_Δ is defined, otherwise $\mathcal{O}(\Delta)$ is undefined.

The sequence

$$(\mathcal{I}_\alpha : \alpha < w)$$

of determinacy pointclasses is defined recursively by iterating the operator $\mathcal{O}$ as long as possible. If $L(\mathbb{R})$ does not satisfy AD^+ , the sequence is empty, otherwise $\mathcal{I}_0$ is set to be the powerset of reals of $L(\mathbb{R})$. If $\mathcal{I}_\alpha$ is defined, then $\mathcal{I}_{\alpha+1}$ is set to be $\mathcal{O}(\mathcal{I}_\alpha)$, if the latter is defined and is a determinacy pointclass. The recursion terminates if the last condition is not met. At the limit stages we take the powerset of reals of the constructible closure of the union of the previous pointclasses. Again, the construction terminates if the obtained pointclass is not a determinacy pointclass. We showed the following theorem.¹⁸

Theorem 2.1 (K, Sargsyan). *Suppose there exists an inner model containing all reals and satisfying $\text{AD}_{\mathbb{R}} +$ “ Θ is regular”. Then for some $\alpha < w$, $L(\mathcal{I}_\alpha)$ is the smallest such model.* $\square$

Even though we actually established a similar fact for theories somewhat beyond $\text{AD}_{\mathbb{R}} +$ “ Θ is regular”, the following problem remains open.

Problem 2.2. *Assuming there exists an inner model containing all reals and satisfying Largest Suslin Axiom¹⁹, does there exist $\alpha < w$ such that $L(\mathcal{I}_\alpha)$ is the smallest such model?* $\square$

¹⁶[Sar15, Main Theorem] ¹⁷[Ste08, proof of Theorem 3.1] ¹⁸[KS24]

¹⁹[ST23, Definition 1.0.1]

I believe I can answer this question in short-term using the techniques already present in the paper of Sargsyan and I.

There exists (provably in ZFC) the $\subseteq$ -smallest inner model $\mathcal{D}$ such that for all $X \in \text{Ord} \cup \{\mathcal{P}(\mathbb{R}) \cap \mathcal{D}\}$, X^ω and $\mu_X \cap \mathcal{D}$ belong to $\mathcal{D}$. Large cardinals imply that this model satisfies AD^+ .²⁰ Whenever the latter is true, u is a successor ordinal and $\mathcal{A}_{u-1}$ is the powerset of reals of $\mathcal{D}$. Thus, this pointclass is the natural limit of the above construction, but it remains unclear how big it is.²¹

Problem 2.3. *Assuming that $\mathcal{D}$ satisfies AD^+ , does it follow (provably) that the minimal model of $\text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$ exists?*²² $\square$

I believe I can answer this question in short-term. The similar question can be asked for other theories as well, for example, Largest Suslin Axiom, but I think I can give an upper bound on how complicated $\mathcal{P}(\mathbb{R}) \cap \mathcal{D}$ can be.

Problem 2.4. *Is it true that every set of reals in $\mathcal{D}$ is (provably) Wadge-reducible to the strategy for a hod mouse with a Woodin limit of Woodin cardinals?*²³ $\square$

I expect this to be a medium- to long-term problem. Finally, I believe that the following is a very ambitious question that can bring a significant insight into the Descriptive Inner Model Problem. I plan to work on it in long-term.

Problem 2.5. *Is there an L -like, bottom-up construction of determinacy pointclasses that produces every determinacy pointclass Δ consisting of universally Baire sets of reals?*²⁴ $\square$

3 SSP Forcibility

One of the fundamental problems in logic is to characterize when a given formula has a model and what that model is. Veličković and I answered a form of this

²⁰Woodin showed (unpublished, cf. [GMS23, Theorem 1.3(2)]) that the existence of a proper class of Woodin cardinals that are limits of Woodin cardinals implies that the so-called *Chang-plus model*, denoted by CM^+ , satisfies AD^+ and for all $X \in \text{CM}^+$, $\mu_X \cap \text{CM}^+$ belongs to the Chang-plus model and is an ultrafilter there. This means that $\mathcal{D} \subseteq \text{CM}^+$ and consequently satisfies AD^+ .

²¹We know that if there exists an inner model that contains it, satisfies $\text{AD}_{\mathbb{R}}$, and has strictly more sets of reals, then in that model, there is a strategy for a hod mouse with a measurable limit of Woodin cardinals (vide [KS24]).

²²Theorem 2.1 implies that if the minimal model for $\text{AD}_{\mathbb{R}} + \text{“}\Theta \text{ is regular”}$ exists, then it is contained $\mathcal{D}$.

²³The results of [GS23] show that this should hold for $L(\text{Ord}^\omega)$ in place of $\mathcal{D}$, while [Ste23a] suggests that if $\mathcal{D}$ is replaced by the Chang-plus model, the conclusion should be true with “measurable Woodin cardinal” in place of “Woodin limit of Woodin cardinals”. The theorem of Woodin of the footnote 20 leads me to believe that this last bound is not optimal and that the optimal one is closer to “Woodin limit of Woodin cardinals”. The model $\mathcal{D}$ is situated in between $L(\text{Ord}^\omega)$ and the Chang-plus model.

²⁴Since divergent models of $\text{AD}_{\mathbb{R}}$ are consistent, we cannot hope to get all determinacy pointclasses in a deterministic manner. Hence, we restrict our attention to the “main line” of determined sets, i.e. the universally Baire sets of reals.

question for infinitary propositional formulas.²⁵ The standard model existence property in this case, called *consistency*, asserts that the formula has a model in some generic extension of the universe. We work with variants of this property, which we call proper, semi-proper, and stationary-set-preserving consistency. A formula is *stationary-set-preserving-consistent* (or *ssp-consistent* for short) if it has a model in the generic extension of the universe via stationary-set-preserving forcing.

The consistency of a formula can be characterized game-theoretically, in terms of the *Model Existence Game*. We provide a variant game $\mathfrak{D}_{\phi, \kappa}^{\text{ssp}}$, depending on the given formula ϕ and an inaccessible cardinal κ , that characterizes the ssp-consistency. More precisely, we prove the following theorem.

Theorem 3.1 (K, Veličković). *Suppose there exists a proper class of inaccessible cardinals. Let ϕ be an infinitary propositional formula. The following are equivalent:*

- a. ϕ is ssp-consistent;
- b. there exists an inaccessible cardinal κ such that Player II has a winning strategy in $\mathfrak{D}_{\phi, \kappa}^{\text{ssp}}$. $\square$

Furthermore, if the formula is ssp-consistent, we give a canonical stationary-set-preserving poset, denoted by $\mathbb{C}_{\phi, \kappa}^{\text{ssp}}$, which forces a model for it. This poset consists of all moves of Player II in $\mathfrak{D}_{\phi, \kappa}^{\text{ssp}}$ according to their winning strategy.

We also give a sufficient condition for ssp consistency in terms of generic elementary embeddings. An infinitary propositional formula ϕ in negation normal form is said to be *AS-good* at an inaccessible cardinal κ if for all stationary subsets S of ω_1 , it holds in $V^{\text{Col}(\omega, < \kappa)}$ that for all models μ for ϕ , there exists an elementary embedding

$$\tau : V \rightarrow W$$

with critical point ω_1^V and a model $\hat{\mu}$ for $\tau(\phi)$ such that $\omega_1^V \in \tau(S)$ and $\hat{\mu} \circ \tau$ coincides with μ on subformulas of ϕ . We showed the following.

Theorem 3.2 (K, Veličković). *Let ϕ be an infinitary propositional formula. If ϕ is AS-good at some inaccessible cardinal $\kappa > \text{rank}(\phi)$, then ϕ is ssp-consistent.* $\square$

We also illustrated how this criterion for ssp consistency can be used to give a somewhat different proof of the fact that MM^{++} implies the axiom $(*)$. These results can be seen as a step towards the problem of forcing $(*)$. A missing step is a way to iterate the procedure of adding a model of an ssp-consistent formula.

Problem 3.3. *Is there a forcing iteration which at every stage chooses an ssp-consistent formula and adds a model to it without killing stationary sets of ω_1 , so that the final universe has witnesses for a large collection of Σ_1 formulas?* $\square$

This problem is short- to medium-term difficulty. Its positive resolution would provide a significant class of ssp posets that can be iterated, thus bringing us one step close to the SSP Iterability Problem.

²⁵Here and elsewhere in this section, I am talking about the paper [KV24].

4 Forcing Iterations and Virtual Models

A *virtual model* is a set that is elementary in its transitive closure and satisfies ZFC^- . A prototypical example of a virtual model is an elementary submodel of some H_θ , and virtual models in general behave much like them. Veličković and I have presented a new way to iterate semiproper posets using these models.^{26,27} In addition to the *working part* which encodes the elements of the posets to be iterated, the conditions of our iteration have a *side-condition part* consisting of virtual models. At the limit stages of the construction we thread virtual models instead of domains of working parts²⁸. We illustrated our method by providing a new proof of the consistency of the saturation of the nonstationary ideal on ω_1 (from the optimal hypothesis).

The importance of this method lies in the fact that it provides an opportunity for generalizations to the iterations of ssp posets. The work that Veličković and I did on the ssp-consistency²⁹ shows that if an iteration of ssp posets can be continued in any way by an ssp poset, then a poset of the form $\mathbb{C}_{\phi, \kappa}^{\text{ssp}}$ (mentioned in the previous section) is one such continuation. This latter poset closely resembles the iterated posets constructed in the iteration using virtual models as side conditions. I believe that many instances of the following problem can be answered in short- to medium-term.

Problem 4.1. *What ssp posets can be iterated using virtual models as side conditions?* $\square$

The answer to this question will probably give an important insight into the SSP Iteration Problem.

5 Local Definability of HOD in $L(\mathbb{R})$

In his seminal 1995 paper, Steel showed that $(\text{HOD} \parallel \Theta)^{L(\mathbb{R})}$ is a mouse, provided that AD^+ holds in $L(\mathbb{R})$. This means that $(\text{HOD} \parallel \Theta)^{L(\mathbb{R})}$ is the core model in itself, but it remained open whether this mouse is a core model in $L(\mathbb{R})$. One of the properties expected of a core model is some kind of inductive (or *local*) definability in their background universe. I verified this for HOD in $L(\mathbb{R})$ in the interval $[\delta_1^2, \Theta)$. More precisely, I proved the following theorem.³⁰

Theorem 5.1 (K.). *Suppose there exist ω Woodin cardinals and a measurable cardinal bigger than them. The following is true in $L(\mathbb{R})$. For all HOD-cardinals η from the interval $[\delta_1^2, \Theta)$, the set $H(\eta^+)^{\text{HOD}}$ is lightface³¹ definable over the structure*

$$L_\chi((H_\eta^{\text{HOD}})^\omega)[\mu_\eta],$$

²⁶[KV23]

²⁷These models are also used to define the game $\mathfrak{D}_{\phi, \kappa}^{\text{ssp}}$ mentioned in the previous section.

²⁸Threading the working parts is the idea behind the revised countable support iteration of [She98].

²⁹[KV24] ³⁰[Kas23]

³¹By this, we mean that only the parameters which are part of the structure are used. Of

where μ_η is the quasi-club filter on $[\eta]^\omega$ and χ is the second ordinal greater than η for which the above structure satisfies $\mathbf{ZF}^- + “\mathcal{P}([\eta]^\omega) \text{ exists}”$. $\square$

course, the additional predicate which is part of the structure is used as well.

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